

Conditional gradient methods. Projected Gradient Descent. Frank-Wolfe Method. Mirror Descent Algorithm Idea.

Seminar

Optimization for ML. Faculty of Computer Science. HSE University

Projection

The **distance** d from point $\mathbf{y} \in \mathbb{R}^n$ to closed set $S \subset \mathbb{R}^n$:

$$d(\mathbf{y}, S, \|\cdot\|) = \inf\{\|x - \mathbf{y}\| \mid x \in S\}$$

We will focus on **Euclidean projection** (other options are possible) of a point $\mathbf{y} \in \mathbb{R}^n$ on set $S \subseteq \mathbb{R}^n$ is a point $\text{proj}_S(\mathbf{y}) \in S$:

$$\text{proj}_S(\mathbf{y}) = \frac{1}{2} \underset{\mathbf{x} \in S}{\text{argmin}} \|x - \mathbf{y}\|_2^2$$

- **Sufficient conditions of existence of a projection.** If $S \subseteq \mathbb{R}^n$ - closed set, then the projection on set S exists for any point.
- **Sufficient conditions of uniqueness of a projection.** If $S \subseteq \mathbb{R}^n$ - closed convex set, then the projection on set S is unique for any point.
- If a set is open, and a point is beyond this set, then its projection on this set does not exist.
- If a point is in set, then its projection is the point itself.

Projection

💡 Bourbaki-Cheney-Goldstein inequality theorem

Let $S \subseteq \mathbb{R}^n$ be closed and convex, $\forall x \in S, y \in \mathbb{R}^n$. Then

$$\langle y - \text{proj}_S(y), x - \text{proj}_S(y) \rangle \leq 0 \quad (1)$$

$$\|x - \text{proj}_S(y)\|^2 + \|y - \text{proj}_S(y)\|^2 \leq \|x - y\|^2 \quad (2)$$

💡 Non-expansive function

A function f is called **non-expansive** if f is L -Lipschitz with $L \leq 1$. That is, for any two points $x, y \in \text{dom} f$,

$$\|f(x) - f(y)\| \leq L\|x - y\|, \text{ where } L \leq 1.$$

It means the distance between the mapped points is possibly smaller than that of the unmapped points.

Non-expansive becomes contractive if $L < 1$.

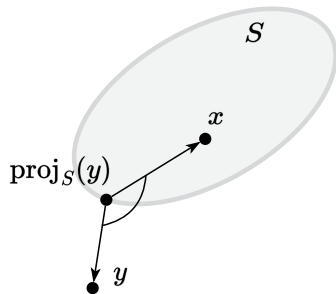


Figure 1: Obtuse or straight angle should be for any point $x \in S$

Problems

i Question

Is projection operator non-expansive?

i Question

Find projection $\text{proj}_S(\mathbf{y})$ onto S , where S :

- l_2 -ball with center 0 and radius 1:

$$S = \{x \in \mathbb{R}^d \mid \|x\|_2^2 = \sum_{i=1}^d x_i^2 \leq 1\}$$

- $\mathbb{R}^d$ -cube:

$$S = \{x \in \mathbb{R}^d \mid a_i \leq x_i \leq b_i\}$$

- Affine constraints:

$$S = \{x \in \mathbb{R}^d \mid Ax = b\}$$

Task 1

Let $\mathcal{L}_L$ denote the set of matrices that are L -Lipschitz:

$$\mathcal{L}_L = \left\{ A \in \mathbb{R}^{m \times n} \mid \|Ax - Ay\|_2 \leq L\|x - y\|_2 \quad \forall x, y \in \mathbb{R}^n \right\}.$$

A projection problem: Given a matrix $M \in \mathbb{R}^{m \times n}$, find the matrix $X \in \mathcal{L}_L$ that is closest to M under a norm:

$$\mathcal{P}_{(L, \zeta)} : \operatorname{argmin}_{X \in \mathcal{L}_L} \frac{1}{2} \|X - M\|_{\zeta}^2.$$

In fact, this problem has a simple solution when $L = 1$ and $\zeta = F$: Can you come up with the idea of solution in such case?

Task 2

Let $\mathcal{S}$ be the nonnegative orthant, find the following projection

$$\text{proj}_{\mathcal{S}}(\mathbf{y}) = \arg \min_{\mathbf{x} \geq 0} \|\mathbf{x} - \mathbf{y}\|_2$$

where $\mathbf{x} \geq 0$ means $\mathbf{x}$ is inside the nonnegative orthant $\mathcal{S} = \{\mathbf{x} \mid x_i \geq 0 \ \forall i\}$.

What if $\mathcal{S} = \{x \mid l \leq X \leq u\}$?

Task 3

Spectraplex is a spectrahedron, defined as the set

$$\mathcal{S} := \{X \in \mathbb{S}_+^n : \text{Tr } X = 1\},$$

Spectraplex = spectra + simplex, meaning eigenvalues in simplex. Spectraplex is the semidefinite analogue of simplex.

Question: given a matrix $Z \in \mathbb{R}^{n \times n}$, what is the projection of Z onto the set $\mathcal{S}$?

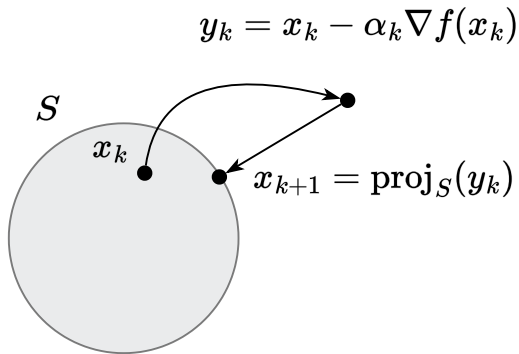
In other words solve the following optimization problem:

$$\arg \min_{X \succeq 0, \text{Tr } X=1} \frac{1}{2} \|X - Z\|_F^2.$$

Projected Gradient Descent (PGD). Idea

$$x_{k+1} = \text{proj}_S(x_k - \alpha_k \nabla f(x_k)) \quad \Leftrightarrow \quad \begin{aligned} y_k &= x_k - \alpha_k \nabla f(x_k) \\ x_{k+1} &= \text{proj}_S(y_k) \end{aligned}$$

Below you can find example of using this method to attack Neural Net: ♣️ Adversarial Attacks.



Frank-Wolfe Method (FWM). Idea

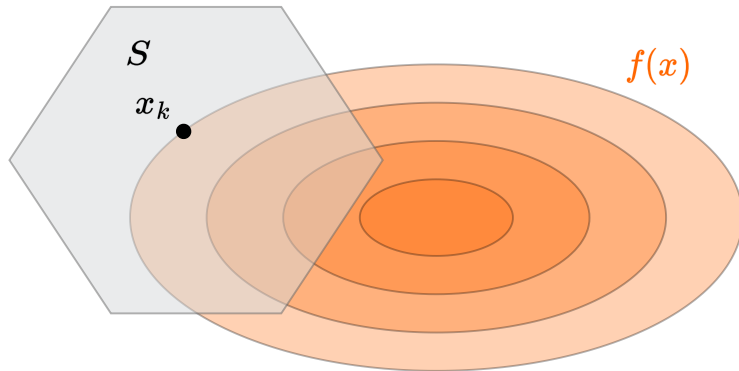


Figure 3: Illustration of Frank-Wolfe (conditional gradient) algorithm

Frank-Wolfe Method (FWM). Idea

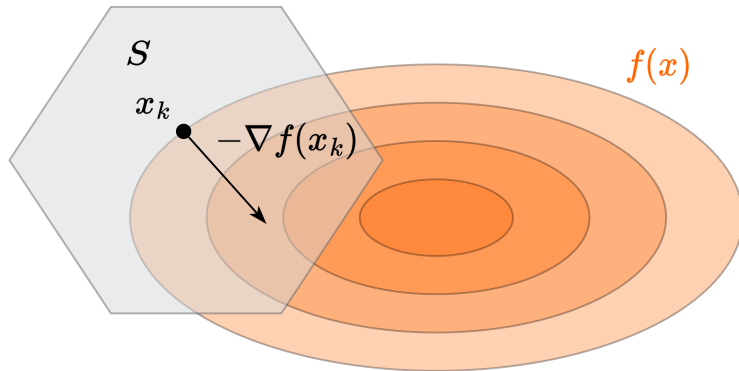


Figure 4: Illustration of Frank-Wolfe (conditional gradient) algorithm

Frank-Wolfe Method (FWM). Idea

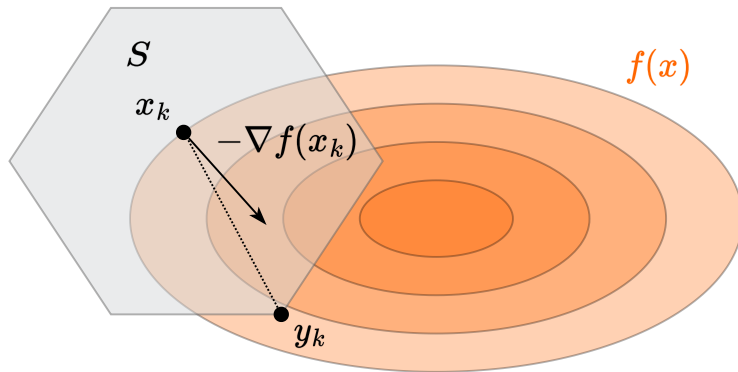


Figure 5: Illustration of Frank-Wolfe (conditional gradient) algorithm

Frank-Wolfe Method (FWM). Idea

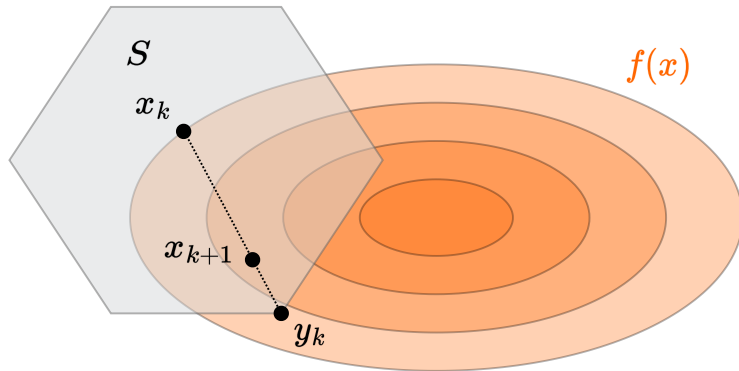


Figure 6: Illustration of Frank-Wolfe (conditional gradient) algorithm

Frank-Wolfe Method (FWM). Idea

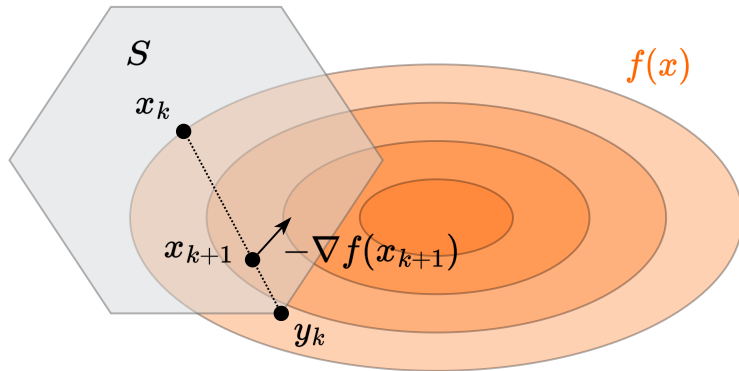


Figure 7: Illustration of Frank-Wolfe (conditional gradient) algorithm

Frank-Wolfe Method (FWM). Idea

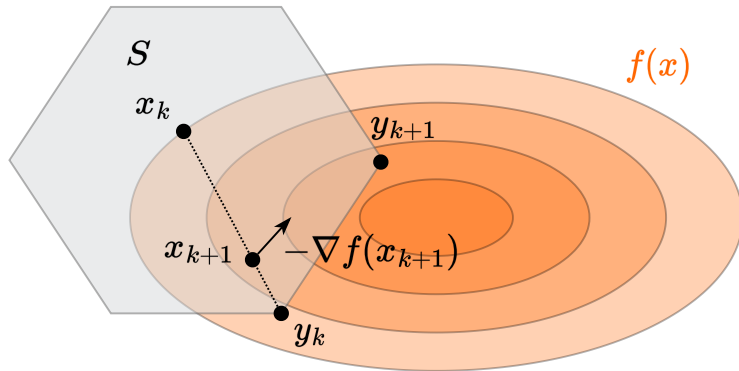


Figure 8: Illustration of Frank-Wolfe (conditional gradient) algorithm

Frank-Wolfe Method (FWM). Idea

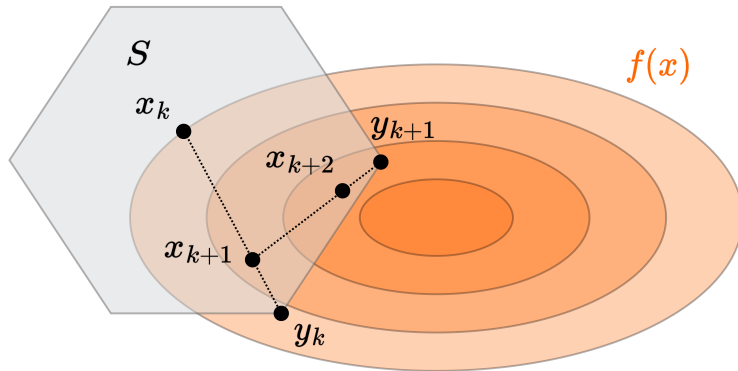


Figure 9: Illustration of Frank-Wolfe (conditional gradient) algorithm

Frank-Wolfe Method (FWM). Idea

$$y_k = \arg \min_{x \in S} f_{x_k}^I(x) = \arg \min_{x \in S} \langle \nabla f(x_k), x \rangle$$

$$x_{k+1} = \gamma_k x_k + (1 - \gamma_k) y_k$$

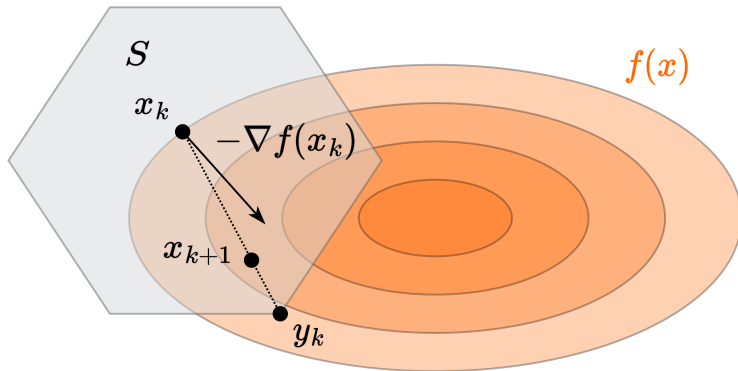


Figure 10: Illustration of Frank-Wolfe (conditional gradient) algorithm

Convergence rate for smooth and convex case

i Theorem

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex and differentiable. Let $S \subseteq \mathbb{R}^n$ be a closed convex set, and assume that there is a minimizer x^* of f over S ; furthermore, suppose that f is smooth over S with parameter L .

- The **Projected Gradient Descent** algorithm with stepsize $\frac{1}{L}$ achieves the following convergence after iteration $k > 0$:

$$f(x_k) - f^* \leq \frac{L\|x_0 - x^*\|_2^2}{2k}$$

- The **Frank-Wolfe Method** achieves the following convergence after iteration $k > 0$:

$$f(x_k) - f^* \leq \frac{2L\|x_0 - x^*\|_2^2}{k+1}$$

💡 FWM specificity

- FWM convergence rate for the μ -strongly convex functions is $\mathcal{O}\left(\frac{1}{k}\right)$
- FWM doesn't work for non-smooth functions. But modifications do.
- FWM works for any norm.

Subgradient method: linear approximation + proximity

Recall SubGD step with sub-gradient g_k :

$$\begin{aligned} x_{k+1} &= \operatorname{argmin}_x \underbrace{f(x_k) + g_k^\top (x - x_k)}_{\text{linear approximation to } f} + \underbrace{\frac{1}{2\alpha} \|x - x_k\|_2^2}_{\text{proximity term}} \\ x_{k+1} = x_k - \alpha_k g_k &\Leftrightarrow \\ &= \operatorname{argmin}_x \alpha g_k^\top x + \frac{1}{2} \|x - x_k\|_2^2 \end{aligned}$$

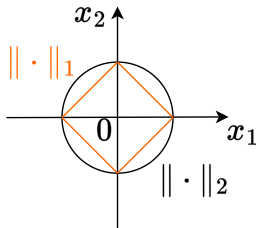


Figure 11: $\|\cdot\|_1$ is not spherical symmetrical

Example. Poor condition

Consider $f(x_1, x_2) = x_1^2 \cdot \frac{1}{100} + x_2^2 \cdot 100$.

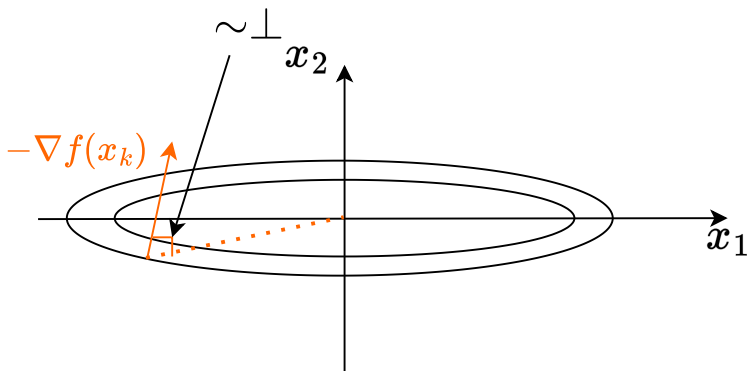


Figure 12: Poorly conditioned problem in $\|\cdot\|_2$ norm

Example. Poor condition

Suppose we are at the point: $x_k = (-10 \quad -0.1)^\top$. SubGD method: $x_{k+1} = x_k - \alpha \nabla f(x_k)$

$$\nabla f(x_k) = \left(\frac{2x_1}{100} \quad 2x_2 \cdot 100 \right)^\top \Big|_{(-10 \quad -0.1)^\top} = \left(-\frac{1}{5} \quad -20 \right)^\top$$

The problem: due to elongation of the level sets the direction of movement $(x_{k+1} - x_k)$ is $\sim \perp (x^* - x_k)$.

The solution: Change proximity term

$$x_{k+1} = \operatorname{argmin}_x \underbrace{f(x_k) + g_k^\top(x - x_k)}_{\text{linear approximation to } f} + \underbrace{\frac{1}{2\alpha}(x - x_k)^\top I(x - x_k)}_{\text{proximity term}}$$

to another

$$x_{k+1} = \operatorname{argmin}_x \underbrace{f(x_k) + g_k^\top(x - x_k)}_{\text{linear approximation to } f} + \underbrace{\frac{1}{2\alpha}(x - x_k)^\top Q(x - x_k)}_{\text{proximity term}},$$

where $Q = \begin{pmatrix} \frac{1}{50} & 0 \\ 0 & 200 \end{pmatrix}$ for this example. And more generally to another function $B_\phi(x, y)$ that measures proximity.

Example. Poor condition

Let's find x_{k+1} for this **new** algorithm

$$\alpha \nabla f(x_k) + \begin{pmatrix} \frac{1}{50} & 0 \\ 0 & 200 \end{pmatrix} (x - x_k) = 0.$$

Solving for x , we get

$$x_{k+1} = x_k - \alpha \begin{pmatrix} 50 & 0 \\ 0 & \frac{1}{200} \end{pmatrix} \nabla f(x_k) = (-10 \ -0.1)^\top - \alpha(-10 \ -0.1)^\top$$

Observation: Changing the proximity term, we **change the direction** $x_{k+1} - x_k$. In other words, if we measure distance using this **new** way, we also **change Lipschitzness**.

Question

What is the Lipschitz constant of f at the point $(1 \ 1)^\top$ for the norm:

$$\|z\|^2 = z^\top \begin{pmatrix} 50 & 0 \\ 0 & \frac{1}{200} \end{pmatrix} z?$$

Example. Robust Regression

Square loss $\|Ax - b\|_2^2$ is very sensitive to outliers.

Instead: $\min \|Ax - b\|_1$. This problem also **convex**.

Let's compute L -Lipshitz constant for $f(x) = \|Ax - b\|_1$:

$$|\|Ax - b\|_1 - \|Ay - b\|_1| \leq L\|x - y\|_2.$$

To simplify calculation: $A = I$, $b = 0$, i.e. $f(x) = \|x\|_1$.

If we take $x = \mathbf{1}_d$, $y = (1 + \varepsilon)\mathbf{1}_d$:

$$|n - (1 + \varepsilon)n| = \varepsilon n \leq L\|x - y\|_2 = \|\varepsilon\|_2 = \sqrt{n\varepsilon^2} = \varepsilon\sqrt{n}.$$

Finally, we get $L = \sqrt{n}$. As we can see, L is **dimension dependent**.

Question

Show that if $\|\nabla f(x)\|_\infty \leq 1$, then $\|\nabla f(x)\|_2 \leq \sqrt{d}$.

References

Examples for the Mirror Descent was taken from the  Lecture.